4 SURVEYING ASTRONOMY

ASTRONOMICAL COORDINATE SYSTEMS

1. The spherical triangle known as the PZS triangle has sides that are segments of three great circles. Which two of those great circles intersect at the star?

(A) the prime vertical and the celestial equator
(B) the hour circle and the horizon
(C) the equinoctial colure and the celestial equator
(D) the hour circle and the vertical circle

Solution: The acronym PZS stands for three points on the celestial sphere: the pole, the zenith, and the star. Each pair of points is included in a great circle. The great circle through the pole and the star is known as the *hour circle*. The great circle through the zenith and the star is known as the *vertical circle*. The great circle through the pole and the zenith is known as the *observer's meridian*.

Answer is (D)

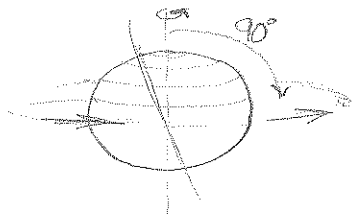
2. Which astronomical coordinates are constantly changing and can readily be measured by a theodolite?

(A) right ascension and declination
(B) hour angle and declination
(C) hour angle and parallactic angle
(D) altitude and azimuth

Solution: The altitude and azimuth of a celestial body can be measured with respect to the observer's horizon and astronomic north. An instrument such as a theodolite can be successfully oriented to these references with relative ease. The references of the other astronomic coordinates include the First Point of Aries and the celestial equator, among others. It would be considerably more difficult to orient a theodolite to such references.

Answer is (D)

3. The angle between an observer's zenith and the celestial equator will always be equal to which of the following?



(A) 90°
(B) the observer's longitude
(C) the right ascension
(D) the observer's latitude

Solution: This problem involves one of the principles of the imaginary system known as the *celestial sphere*. Terrestrial positions are transferred to the celestial sphere by lines prolonged from the center of the earth to the surface of the celestial sphere. Therefore, the angular distance from the terrestrial equator to an observer's position on the earth, which is the observer's latitude, remains unchanged when the points between which the latitude is reckoned are transferred to the surface of the celestial sphere.

Answer is (D)

4. One sidereal hour after a star has set exactly in the west, the First Point of Aries stands precisely at upper transit on an observer's meridian. At that instant, what is the right ascension of the star?

(A) 255°00′
(B) 180°00′
(C) 105°30′
(D) 330°15′

Solution: *Sidereal time*, also known as *star time*, is the system used to measure the movement of the stars. While this system of time is not synchronous with solar time, there are nevertheless 24 sidereal hours in a sidereal day. Zero hour in local sidereal time occurs when the First Point of Aries, a point on the celestial equator, is at upper transit on an observer's meridian. Twenty-four sidereal hours later, the First Point of Aries once again stands at the same meridian. Since it must travel 360° in those 24 sidereal hours, it must travel 15° per sidereal hour.

The star that sets precisely in the west must also be on the celestial equator. As with all stars, its right ascension is measured from the First Point of Aries in the direction opposite the apparent rotation of the celestial sphere. Had the star in question set at the same moment that the First Point of Aries reached upper transit, its right ascension would be 270° (remember, right ascension is measured opposite the direction of apparent rotation). However, the star set one sidereal hour, or 15°, before the First Point of Aries reached upper transit. Therefore, the star's right ascension is 270° less 15°, or 255°.

Answer is (A)

5. If your latitude was 39°47′30″N, what would be the altitude of the sun at apparent noon on the day of the summer solstice?

(A) $50°17\frac{1}{2}'$
(B) $73°39'$
(C) $23°26\frac{1}{2}'$
(D) $39°47\frac{1}{2}'$

Solution: The sun is on an observer's meridian at upper transit each day at what is known as apparent noon for that particular location. The sun's declination, that is, its angular distance from the celestial equator, changes throughout the year. Its declination is at its maximum in the northern hemisphere on the summer solstice, which occurs around June 22 each year. The sun's declination on the summer solstice is about +23°26′30″.

The angular distance from the celestial equator to the observer's zenith is 39°47′30″ (the observer's latitude).

The angular distance from the observer's horizon to the observer's zenith is 90°00′00″.

The angular distance from the observer's horizon to the celestial equator is $90°00'00'' - 39°47'30'' = 50°12'30''$.

The declination of the sun, its angular distance from the celestial equator, on the summer solstice is $23°26'30''$.

The altitude of the sun at apparent noon on the summer solstice for an observer at latitude $39°47'30''$N can be found by addition.

angular distance from the celestial equator to the observer's horizon	=	$50°12'30''$
angular distance from the celestial equator to the sun on the summer solstice	=	$\underline{+23°26'30''}$
altitude of the sun on the summer solstice for an observer at latitude $39°37'30''$N	=	$73°39'00''$

Answer is (B)

6. What is the latitude of the place where the maximum altitude of the sun above the horizon achieved at any time of the year is $23°26\frac{1}{2}'$?

(A) $90°00'00''$S
(B) $23°26'30''$N
(C) $90°00'00''$N
(D) both A and C

Solution: Since the maximum declination of the sun is $23°26\frac{1}{2}'$ north and south of the celestial equator, any latitude less than 90° must witness an altitude greater than $23°26\frac{1}{2}'$ at some time. The only place where the maximum altitude of the sun is the same as its maximum declination is where the celestial equator corresponds with the horizon—at one of the earth's poles.

Answer is (D)

7. What is the declination of a star if that star passes through your zenith where your latitude is 35°N?

(A) 35°N
(B) 55°N
(C) 90°N
(D) 35°S

Solution: The angle between the celestial equator and the zenith of the observer is the latitude of the observer. Since declination of stars is also measured from the celestial equator along the stars' hour circles, the star in question must have a declination equal to that of the observer's latitude. Any star that passes through an observer's zenith must have a declination equal to the observer's latitude. This principle is the foundation of one technique for determining the latitude of a position.

Answer is (A)

8. Standing at 40°N latitude, you observe the sun rise precisely in the east. What is the declination of the sun and what is the approximate date?

(A) declination 0°, June 22 or Dec. 22
(B) declination $23°26\frac{1}{2}'$, June 22 or Dec. 22
(C) declination 0°, Mar. 21 or Sept. 23
(D) declination $23°26\frac{1}{2}'$, Mar. 21 or Sept. 23

Solution: In order to rise exactly in the east, the sun must lie directly on the celestial equator. This occurs twice a year, on Mar. 21 (the vernal equinox) and Sept. 23 (the autumnal equinox). These dates vary somewhat from year to year because the exact times of the equinoxes vary.

Answer is (C)

9. Concerning the PZS triangle, colatitude is to latitude as polar distance is to which of the following?

(A) declination
(B) altitude
(C) hour angle
(D) zenith distance

Solution: The latitude of an observer is the complement of the side of the PZS triangle called the *colatitude*. The complement of the polar distance is the declination of a star.

Answer is (A)

10. Which value corresponds to the sum of a star's hour angle and its right ascension?

(A) the azimuth of the star
(B) the longitude of the observer
(C) the zenith distance of the star
(D) the sidereal time at the observer's position

Solution: The right ascension of a star is measured in the opposite direction from the apparent rotation of the celestial sphere. Hour angles are measured in the same direction as the apparent rotation of the celestial sphere. Both values are measured to the hour circle of the star, but from different origins. The right ascension is from the First Point of Aries and the local hour angle is from the observer's meridian. Their sum is equal to the local sidereal time at the observer's position, since zero hour sidereal would have occurred when the First Point of Aries was last at the upper transit on the observer's meridian.

Answer is (D)

GEOGRAPHICAL COORDINATES AND THEIR ELEMENTS

11. Which of the following terms describes a great circle on which every point is equidistant from the north and the south pole?

(A) the prime meridian
(B) the hour circle
(C) the Tropic of Capricorn
(D) the terrestrial equator

Solution: The terrestrial equator lies on a plane that passes through the earth's center and is perpendicular to the polar axis of the earth. These two properties assure that every point on the terrestrial equator must be equidistant from the two poles of the earth.

Answer is (D)

12. Which of the following statements correctly describes properties of the geographical coordinate known as *latitude*?

(A) Latitude is measured in degrees, minutes, and seconds north and south of the equator.
(B) Every position on the earth has a unique latitude, which is unlike the latitude of any other position on the earth.
(C) A parallel of latitude is a great circle on the surface of the earth.
(D) All of the above are true.

Solution: A parallel of latitude is a small circle on the surface of the earth. Every point along a parallel of latitude has the same latitude, so it cannot be said that any position, except perhaps the north and south poles, has a latitude unique to itself. Latitude is measured in degrees, minutes, and seconds north and south of the equator.

Answer is (A)

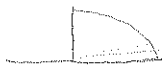
13. A second of latitude near the equator bears what relationship to a second of latitude near one of the poles?

(A) A second of latitude near the poles is approximately a foot longer than a second of latitude near the equator.

(B) A second of latitude has the same length in feet regardless of its distance from the equator.

(C) A second of latitude near the poles is approximately a foot shorter than a second of latitude near the equator.

(D) Each second of latitude north of the equator grows progressively shorter until finally it reaches a point at the pole.



Solution: A second of latitude near the poles is approximately a foot longer than a second of latitude near the equator. A degree of latitude near the equator is approximately 68.7 mi, while a degree of latitude near the poles is approximately 69.4 mi. The length of a second can be approximated by dividing these quantities by 3600—the number of seconds in a degree—and multiplying the answer by 5280 ft.

$$\frac{68.7 \text{ mi}}{3600 \text{ sec}} = 0.0191$$

$$(0.0191)(5280 \text{ ft}) = 100.76 \text{ ft}$$

$$\frac{69.4 \text{ mi}}{3600 \text{ sec}} = 0.0192$$

$$(0.0192)(5280 \text{ ft}) = 101.79 \text{ ft}$$

The debate over the length of a degree of latitude measured along a meridian became so intense in 1735 that the French Academy of Sciences sent geodetic surveying expeditions to settle the issue. Expeditions went to Peru and Lapland to take their measurements. The expeditions discovered that the earth is flattened at the poles and determined that a degree of latitude near the poles is longer. The earth is an oblate spheroid.

Answer is (A)

14. Which of the following statements correctly describes properties of the geographical coordinate known as *longitude*?

(A) Each meridian of longitude lies in the plane of a great circle.
(B) The distance along a parallel of latitude through a degree of longitude grows smaller as the latitude approaches 90°.
(C) 15° of longitude equals one mean solar hour.
(D) All of the above are true.

Solution: The plane of each meridian of longitude passes through the center of the earth. Therefore, each meridian is an arc of a great circle. The distance between meridians along a parallel of latitude grows quite short near the poles.

Since there are 24 hours in a mean solar day and 360° of longitude around the earth, 15° equals one mean solar hour.

$$\frac{360^\circ}{24 \text{ hr}} = 15^\circ$$

Answer is (D)

15. The longitude of a place is 100°15′32″. What would this longitude be if it were expressed in mean solar hours, minutes, and seconds from Greenwich?

(A) 6 hr 40 min 37 sec
(B) 6 hr 41 min 02 sec
(C) 6 hr 01 min 02 sec
(D) 6 hr 55 min 45 sec

Solution: Greenwich has been the prime, or zero, meridian since 1884. Each 15° of longitude is one mean solar hour. Since Greenwich's longitude is zero, the answer may be found by dividing the given longitude by 15.

$$\frac{100^\circ 15' 32''}{15} = 6 \text{ hr } 41 \text{ min } 02 \text{ sec}$$

Answer is (B)

16. Which of the following Greek letters is the standard symbol for longitude?

(A) β
(B) ς
(C) ϕ
(D) λ ← LAMBDA, LONGITUDE

Solution: The standard notation for longitude is the lowercase Greek letter lambda, or λ. The abbreviation Wλ is used to indicate west longitude and Eλ is used for east longitude.

Answer is (D)

17. Which of the following Greek letters is the standard symbol for latitude?

(A) θ
(B) ϕ ← PHI, LATITUDE
(C) ψ
(D) μ

Solution: The standard notation for latitude is the lowercase Greek letter phi, or ϕ. Nϕ indicates north latitude and Sϕ indicates south latitude. North latitude is sometimes called *positive* (+), and south latitude is called *negative* (−).

Answer is (B)

THE PZS TRIANGLE

18. Which of the following correctly identifies the points of intersection of the three great circles that form the spherical triangle called the *PZS triangle*?

(A) P: parallactic
Z: zone
S: south

(B) P: pole
Z: zero
S: station

(C) P: prime meridian
Z: zenith
S: Sirius

(D) P: pole
Z: zenith
S: star

Solution: The PZS triangle is formed by the arcs of three great circles that intersect at the pole, the observer's zenith, and the star. The pole may be either the north or south, and the star may be any celestial body.

Answer is (D)

19. The angle between the great circles that form two of the sides of the PZS triangle is the azimuth of the star being observed. Which are those two circles?

(A) the prime vertical and the hour circle through the star
(B) the observer's meridian and the hour circle through the star
(C) the vertical circle through the star and the observer's meridian
(D) the vertical circle through the star and its hour circle

Solution: The PZS triangle has six elements, one of which is called the *zenith angle*. The zenith angle is the angle between the vertical circle through the star and the observer's meridian. Both the vertical circle through the star and the observer's meridian are thought of as great circles on the celestial sphere.

Answer is (C)

20. Which of the following arcs of the great circles, both complements of the sides of the PZS triangle, are used to find the azimuth angle of a star using the hour angle method?

(A) latitude and declination
(B) zenith distance and declination
(C) coaltitude and latitude
(D) polar distance and zenith distance

Solution: The three sides of the PZS triangle are segments of three great circles. These segments are known as the *colatitude*, the *polar distance*, and the *zenith distance*. The polar distance is also known as the *codeclination* and the zenith distance is also known as the *coaltitude*.

The two sides of the PZS triangle whose complements are critical to solving the azimuth of a star by the hour angle method are the colatitude and the polar distance. The complements are the latitude of the observer and the declination of the star.

The formula for the hour angle is

$$\tan z = \frac{\sin t}{\cos\phi \tan\delta \cos t}$$

The latitude (complement of the colatitude) and the declination (complement of the polar distance) are both essential to the formula's solution.

Answer is (A)

21. Which element of the PZS triangle is the least used in surveying astronomy?

(A) the zenith angle
(B) the hour angle
(C) the parallactic angle
(D) the colatitude

Solution: The parallactic angle, formed by the intersection of the vertical circle and the hour circle, is the least-used element of the PZS triangle. While it

certainly can be calculated, the angle is not nearly as useful as the hour angle or the zenith angle, the other two angles in the PZS triangle.

Answer is (C)

22. When a star is at upper transit on the observer's meridian, which statement concerning its PZS triangle is correct?

(A) The hour angle is 90°.
(B) The hour circle is coincident with the celestial equator.
(C) The colatitude is equal to the coaltitude.
(D) It is coincident with the observer's meridian and is a straight line.

Solution: Two of the three constituent points of the PZS triangle, the observer's zenith and the pole, always lie on the observer's meridian. Most of the time the third point, the star, does not. However, when the star reaches upper transit on the observer's meridian, the PZS triangle, from the observer's point of view, is reduced to a straight line that is coincident with the observer's meridian.

Answer is (D)

THE EARTH'S ORBIT AND THE ECLIPTIC

23. Which of the following defines the ecliptic?

(A) the sun's apparent yearly path
(B) the intersection of the earth's orbital plane with the celestial sphere
(C) the name of the point in the earth's orbit where it is farthest from the sun
(D) both A and B

Solution: The earth's yearly orbit around the sun creates the apparent movement of the sun through the sky. The apparent path of the sun traced out on the celestial sphere is known as the *ecliptic*.

Answer is (D)

24. What are the names given to the two points of intersection between the celestial equator and the ecliptic?

(A) the summer solstice and the winter solstice
(B) the vernal equinox and the autumnal equinox
(C) the solstitial points
(D) the Tropic of Cancer and the Tropic of Capricorn

Solution: The two equinoctial points, where the ecliptic intersects the celestial equator, indicate the position of the sun when the hours of daylight and the hours of darkness are approximately equal.

Answer is (B)

25. Which of the following symbols is used to indicate the vernal equinox?

(A) Υ
(B) ζ
(C) Ψ
(D) α

Solution: The vernal equinox is also known as the First Point of Aries. It is symbolized by the uppercase Greek upsilon (or the ram's horn), Υ.

Answer is (A)

26. Which of the following angles most closely indicates the value known as the obliquity of the ecliptic?

(A) $16°56'45''$
(B) $59°43'12''$
(C) $74°12'30''$
(D) $23°26'30''$

Solution: The axis of rotation of the earth is not perpendicular to the plane of its orbit around the sun. The axis is tilted with respect to the orbital plane

by an angle of $66°33'30''$. Therefore, the acute angle between the ecliptic and the celestial equator, known as the *obliquity of the ecliptic*, is the complement of $66°33'30''$, or $23°26'30''$.

Answer is (D)

27. Moving along its yearly path, the sun crosses the celestial equator in an apparent motion from north to south in September. This position of the sun has more than one name. Which of the following names apply to the position?

(A) the vernal equinox and the First Point of Aries
(B) the summer solstice and the First Point of Taurus
(C) the autumnal equinox and the First Point of Libra
(D) the winter solstice and the First Point of Scorpio

Solution: The autumnal equinox occurs about Sept. 23 each year when the sun's apparent motion carries it from the northern to the southern hemisphere. The position on the celestial equator where this crossing occurs is called the *First Point of Libra.*

Answer is (C)

28. The orbit of the earth around the sun follows a path that quite closely resembles the shape of what geometric figure defined in Kepler's laws of planetary motion?

(A) a cycloid
(B) a hyperbolic paraboloid
(C) a cardioid
(D) an ellipse

Solution: The first of Kepler's laws states that every planet orbits the sun in an ellipse, with the sun itself at one focus of that ellipse. Observation has since proven this assertion fundamentally correct. The path of the earth around the sun, though nearly a circle, is elliptical.

Answer is (D)

29. A planet's positions at its closest approach and farthest distance from the sun have been given what names?

(A) apogee: the position closest to the sun
perigee: the position farthest from the sun
(B) aphelion: the position farthest from the sun
perihelion: the position closest to the sun
(C) apogee: the position farthest from the sun
perigee: the position closest to the sun
(D) aphelion: the position closest to the sun
perihelion: the position farthest from the sun

Solution: The terms *apogee* and *perigee* refer to the farthest and closest positions, respectively, of celestial objects relative to the earth. The terms *aphelion* and *perihelion* apply to the farthest and closest positions, respectively, of objects relative to the sun.

Answer is (B)

30. When is the earth nearest to and farthest from the sun?

(A) The earth is nearest to the sun about Jan. 4 of each year.
(B) The earth is farthest from the sun about July 6 of each year.
(C) The earth is nearest to the sun about Mar. 21 of each year.
(D) Both A and B are true.

Solution: Unlikely as it seems, the earth is closest to the sun in January and farthest from it in July. The difference in the warmth of these months in the northern hemisphere is explained by the tilt of the earth's axis with respect to its orbital plane, rather than by its distance from the sun.

Answer is (D)

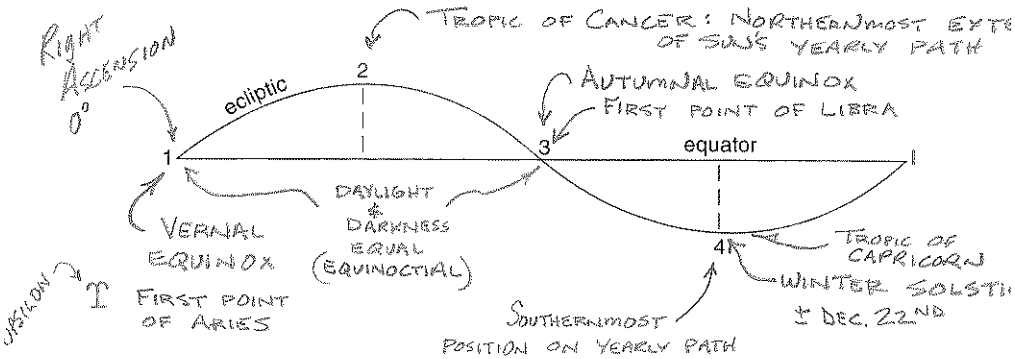
31. Approximately how many months separate the summer solstice from the autumnal equinox?

(A) 6 months
(B) 3 months
(C) 7 months
(D) 1 month

Solution: The summer solstice occurs about June 22 each year; the autumnal equinox occurs about Sept. 23. The period between the autumnal equinox and the summer solstice is approximately 3 months.

Answer is (B)

The following three problems refer to the illustration of the yearly path of the sun along the ecliptic. Note that the position labeled 1 is the First Point of Aries.



32. Which of the labeled positions would be found on the Tropic of Cancer and the Tropic of Capricorn, respectively?

(A) Tropic of Cancer: 2
Tropic of Capricorn: 4
(B) Tropic of Cancer: 4
Tropic of Capricorn: 2
(C) Tropic of Cancer: 1
Tropic of Capricorn: 3
(D) Tropic of Cancer: 2
Tropic of Capricorn: 3

Solution: The Tropic of Cancer is the parallel of latitude that is defined by the most northerly extent of the sun's yearly path; the Tropic of Capricorn is similarly defined by its most southerly extent.

Answer is (A)

33. What angular values of right ascension and declination would be correct for the sun at the moment of the autumnal equinox?

(A) $\alpha = 0°$
$\delta = 90°$

(B) $\alpha = 90°$
$\delta = 180°$

(C) $\alpha = 0°$
$\delta = 180°$

(D) $\alpha = 180°$
$\delta = 0°$

Solution: The autumnal equinox occurs at the position labeled 3. The equinox occurs when the sun is on the equator and the declination is 0°. Since declination is measured from the equator, it must be zero. Right ascension is measured from the First Point of Aries. It is measured along the equator, opposite the direction of apparent rotation of the celestial sphere. The autumnal equinox occurs opposite or 180° from the vernal equinox, which occurs at the position labeled 1.

Answer is (D)

34. Which number indicates the position of the sun at the winter solstice?

(A) 1
(B) 2
(C) 3
(D) 4

Solution: The position labeled 4 indicates the most southerly position of the sun on its yearly path. The winter solstice occurs when the sun is in that position, about Dec. 22 of each year.

Answer is (D)

35. Which of the following best describes the meaning of the term *geoid*?

(A) the figure generated by revolving an ellipse about its minor axis
(B) a surface that is everywhere perpendicular to the direction of gravity
(C) a gently undulating surface that departs from a true spheroidal surface by amounts up to about 300 ft
(D) both B and C

Solution: The geoid is the reference surface for astronomical observations. Its surface is called *equipotential* due to its property of being always perpendicular to the direction of gravity.

Answer is (D)

SOLAR OBSERVATION FOR AZIMUTH

36. When working with an observation to the sun's center that was made to one of its limbs, a correction based on its semidiameter is used. Which statement correctly characterizes the variation in that correction?

(A) It will grow larger as the altitude of the sun increases.
(B) It will grow smaller as the altitude of the sun increases.
(C) It will frequently exceed $00°16\frac{1}{2}'$.
(D) Both A and C are true.

Solution: The angular distance from the center of the sun to one of its limbs is known as its semidiameter. From our point of view on earth, the sun is never more than $00°16\frac{1}{2}'$ from center to limb, but the angular correction under consideration is nearly always greater.

The mathematical expression used to calculate the angular correction, dH, applied to an observation of one of the sun's limbs to correct it to the center is

$$dH = \frac{\text{the sun's semidiameter}}{\cos h}$$

h is the altitude of the sun. Two calculations illustrate the effect that altitude plays in the size of the correction.

$$\begin{aligned} dH &= \frac{00°16'30''}{\cos 45°} \\ &= 00°23'20'' \end{aligned}$$

$$\begin{aligned} dH &= \frac{00°16'30''}{\cos 50°} \\ &= 00°25'40'' \end{aligned}$$

As the altitude of the sun increases, the dH correction increases.

Answer is (D)

37. The true sun does not travel across the sky at a constant rate, but the mean sun does. Which statement is true of the relationship between the two?

(A) Coordinated Universal Time (UTC), or Zulu time, more nearly corresponds to the movement of the mean sun than that of the true sun.
(B) The true sun moves along the ecliptic, while the mean sun is imagined to move along the celestial equator.
(C) The difference between the true sun's and the mean sun's positions can be found using the equation of time.
(D) All of the above are true.

Solution: The mean sun is a product of human imagination. Its constant motion along the celestial equator is in contrast to the true sun's irregular motion along the ecliptic, making the mean sun a much better timekeeper than the true sun. The relationship between the two suns is known as the *equation of time*, which varies from 0 to 16 min.

Answer is (D)

38. The local hour angle of the sun can be found using which formula?

(A) GHA − west longitude
(B) SHA + local sidereal time
(C) GHA + east longitude
(D) both A and C

Solution: The Greenwich hour angle (GHA) is measured from the prime meridian in the direction of the apparent rotation of the celestial sphere, which is east to west. This corresponds to the direction of measurement of west longitude. However, the measurement of the GHA counters the direction of the measurement of east longitude; therefore, east longitude is added to the GHA to produce the local hour angle (LHA).

Answer is (D)

39. Which small circles on the earth's surface have latitudes that correspond to the maximum declination of the sun?

(A) the Tropics of Cancer and Capricorn
(B) the Arctic and Antarctic circles
(C) the parallel of declination of Sirius
(D) the equinoctial colures

Solution: Due to the tilt of the earth's axis, the maximum declination reached by the sun each year is $23°26'30''$. In other words, at some time during the year the sun is directly overhead between the latitudes of $23°26'30''$N and $23°26'30''$S. These lines of latitude have been given the names Tropic of Cancer and Tropic of Capricorn, respectively.

Answer is (A)

40. During an observation to determine the altitude of the sun, which of the following variables must be corrected for?

(A) the temperature of the air
(B) the pressure of the air
(C) the azimuth of the sun
(D) both A and B

Solution: The apparent altitude of a celestial body is greater than its true altitude due to the refraction of light as it passes through the earth's atmosphere. In an attempt to quantify that refraction, adjustments are made to the observed altitude. These adjustments are based on an approximation of the density of the air at the time of the observation made by measuring the air temperature and pressure.

Answer is (D)

41. What significance does the following formula have in preparing a solar observation?

$$\text{UTC} + \text{DUT1} = \text{UT1}$$

(A) The calculation has no significance to a solar observation.
(B) The calculation corrects UTC for significance to tenths of a second.
(C) The calculation applies to sidereal time rather than solar time.
(D) The calculation is the correction of Zulu time to reflect time intervals measured by the true sun rather than the mean sun.

Solution: DUT1 is a correction expressed in tenths of a second that can be determined from a series of double ticks broadcast on radio station WWV by the National Bureau of Standards. This correction is applied to UTC to significantly improve the accuracy of solar calculations.

Answer is (B)

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42. What would be the right ascension of the sun at the moment of the vernal equinox?

(A) 0°
(B) 90°
(C) 180°
(D) 270°

Solution: The right ascension of a celestial body is measured from the First Point of Aries, which is an arbitrary point on the celestial equator. The sun would coincide with that position at the moment of the vernal equinox.

Answer is (A)

43. Which of the following correctly identifies the symbols in the hour angle formula?

$$\tan z = \frac{\sin t}{\cos\phi \tan\delta \cos t}$$

(A) z = azimuth
ϕ = longitude
δ = latitude
t = GHA

(B) z = zenith angle
ϕ = latitude
δ = declination
t = LHA

(C) z = zenith distance
ϕ = declination
δ = longitude
t = right ascension

(D) z = hour angle
ϕ = latitude
δ = longitude
t = LHA

Solution: The use of Greek letters in surveying astronomy is fairly consistent. Ordinarily, ϕ stands for latitude, δ stands for declination, and so on.

Answer is (B)

44. The right ascension of so-called fixed stars is a virtually constant value for each star. The sun is not fixed, and its right ascension is constantly changing. Which other astronomical coordinate is virtually constant for fixed stars but not for the sun?

(A) azimuth
(B) altitude
(C) hour angle
(D) declination

Solution: The declination of a fixed star, like right ascension, remains virtually constant throughout the year. Because it is so close to the earth, the same values for the sun change constantly.

Answer is (D)

45. Which of the following defines *obliquity of the ecliptic*?

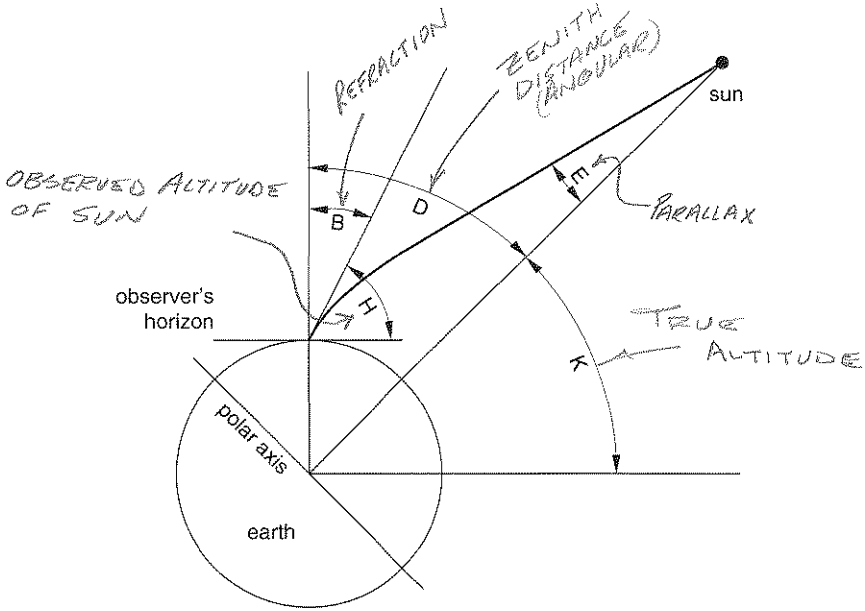
(A) the precession and nutation of the earth's rotation
(B) the tidal force exerted on the earth by the moon
(C) the difference between the sidereal day and the solar day
(D) the angle of the earth's rotational axis with respect to the orbital plane of the earth around the sun

Solution: Among the consequences of the obliquity of the ecliptic are the seasons of the year and the Tropics of Cancer and Capricorn. The earth's rotational axis differs by approximately $23°26\frac{1}{2}'$ from a perpendicular to the orbital plane created by the earth and sun.

Answer is (D)

PARALLAX, REFRACTION, AND THE SEMIDIAMETER OF THE SUN

The following five problems refer to the illustration shown.



46. Which of the following terms best defines the effect indicated by the letter E in the illustration?

(A) parallax
(B) refraction
(C) halation
(D) diffusion

Solution: Parallax is generally the shift in the apparent position of an observed object due to the shift of the position of the observer. As used in surveying astronomy, parallax refers to the difference between the apparent position of a celestial body as it would be seen from the earth's center and the body as seen from the observer's location.

Answer is (A)

47. Which of the following terms best defines the value indicated by the letter H in the illustration?

(A) refraction
(B) the true altitude of the sun
(C) the zenith distance of the sun
(D) the observed altitude of the sun

Solution: The altitude of a celestial body is measured from the observer's horizon to the apparent position of the body. However, the body consistently appears to be higher in the sky than it actually is due to the density of the atmosphere through which the observation must be made. Therefore, there is a difference between the observed altitude of a celestial body and its true altitude.

Answer is (D)

48. Which of the following terms best defines the effect indicated by the letter B in the illustration?

(A) refraction
(B) halation
(C) diffusion
(D) parallactic angle

Solution: When light from a celestial body passes through the atmosphere of the earth, it bends toward the surface of the earth. This refraction causes the body to appear higher in the sky than it actually is.

Answer is (A)

49. Which of the following terms best defines the value indicated by the letter D in the illustration?

(A) the polar distance of the sun
(B) the zenith distance of the sun
(C) the codeclination of the sun
(D) the declination of the sun

Solution: The angular distance from the zenith of the observer to the celestial body is the zenith distance.

Answer is (B)

50. Which of the following terms best defines the value indicated by the letter K in the illustration?

(A) the declination of the sun
(B) the coaltitude of the sun
(C) the true altitude of the sun
(D) the codeclination of the sun

Solution: The center of the earth is assumed to be the center of the imaginary celestial sphere. The true altitude of celestial bodies is the altitude that would be measured from the center of the earth without the effects of refraction or parallax, if such a thing were possible.

Answer is (C)

51. It is sometimes said that parallax has no effect on the observed altitudes of fixed stars. Which of the following statements most correctly explains this idea?

(A) Observations of the altitude of the sun, moon, or planets in the solar system are subject to distortion as light passes through the earth's atmosphere. Similar observations of fixed stars are not subject to such distortion and therefore require no correction for parallax.

(B) Parallax is the difference between the direction of a celestial body as seen from the earth's center and as seen from an observer's point of view on the earth's surface. Strictly speaking, this difference must exist in all astronomical observations, but in the case of fixed stars, the effect is so small as to be immeasurable.

(C) The distance from the earth to the celestial body under observation affects parallax.

(D) Both B and C are true.

Solution: Parallax is the difference in the direction of a celestial body as seen from the earth's center and from an observer's location on earth. Bodies in our own solar system are much closer to earth than even the nearest fixed stars; as a consequence, the effect of parallax is much larger when taking observations on the sun, moon, and planets than on fixed stars. For all practical purposes, the altitude of even the nearest fixed star is exactly the same no matter where on earth the observation is made.

Answer is (D)

52. At what position of the sun would an observer find the effect of parallax to be zero?

(A) at the observer's horizon

(B) at the observer's zenith

(C) at an altitude of 45°

(D) at an altitude of 23°26′30″

Solution: As the altitude of the sun increases, the effect of parallax diminishes. If the sun reaches the observer's zenith, the effect of parallax is zero.

Answer is (B)

53. The sun's parallax, which is always less than 10 sec of arc, is sometimes applied to an observed altitude that has been corrected for the effect of refraction. How is this correction applied?

(A) It is always added.
(B) It is always subtracted.
(C) It is sometimes added and sometimes subtracted.
(D) It is multiplied by the observed altitude.

Solution: The correction for parallax is always added to an observed altitude. The effect of parallax is to reduce the altitude that would be measured at the earth's center. Therefore, when calculating the true altitude, which is the observed altitude at the earth's center, parallax must be added.

Answer is (A)

54. Which of the following statements correctly describes the effect of refraction on the observation of the altitude of celestial bodies?

(A) Observed altitudes of the sun, moon, and planets must be corrected for the effect of refraction in order to find their true altitudes, but observed altitudes of fixed stars need not be corrected.
(B) As with the sun, moon and planets, the observed altitudes of fixed stars must be corrected for the effect of refraction in order to find their true altitudes.
(C) The corrections applied to an observed altitude to compensate for the effects of parallax and refraction are equal.
(D) Refraction decreases with the density of the atmosphere.

Solution: Light from a fixed star is refracted before reaching an observer on earth just as light from the sun is. The denser the atmosphere, the larger the correction that must be applied to the observed altitude to find the true altitude.

Answer is (B)

55. The effect of refraction causes a celestial body to appear higher in the sky than its true altitude. Which of the following statements correctly describes consequences of this fact?

(A) The sun can appear to be just above the observer's horizon when it is actually entirely below it.

(B) A fixed star that is apparently just above an observer's horizon may, in fact, be half a degree below it.

(C) Refraction is zero for a celestial body at the observer's zenith.

(D) All of the above are true.

Solution: The effect of refraction is greatest near the horizon, where the observer must see the celestial body through the thickest layer of atmosphere. Refraction near the horizon reaches nearly $00°33'$. The effect of refraction diminishes as the altitude of a celestial body increases, and it reaches zero at the observer's zenith.

Answer is (D)

56. Which of the following statements correctly describes the relationship between variations in temperature and barometric pressure and variations in the effect of refraction?

(A) As the air temperature and barometric pressure increase, the effect of refraction increases.

(B) As the air temperature increases and the barometric pressure decreases, the effect of refraction increases.

(C) As the air temperature decreases and the barometric pressure increases, the effect of refraction increases.

(D) As the air temperature and barometric pressure decrease, the effect of refraction increases.

Solution: As the atmosphere through which an observer sees a celestial body becomes denser, the effect of refraction increases. A decrease in air temperature causes the air to become more dense, as does an increase in barometric pressure.

Answer is (C)

57. An observed altitude of the sun is $35°15'42''$, the refraction correction is $00°01'24''$, and the parallax correction is $00°00'07''$. What is the true altitude of the sun?

(A) $35°14'11''$
(B) $35°14'25''$
(C) $35°16'59''$
(D) $35°17'13''$

Solution: The true altitude of the sun can be derived from the observed altitude using the following expression.

$$\begin{aligned}\text{true altitude} &= \text{measured altitude} - \text{refraction} + \text{parallax}\\ &= 35°15'42'' - 00°01'24'' + 00°00'07''\\ &= 35°14'25''\end{aligned}$$

Answer is (B)

58. In preparation for an observation on the star Sirius, a true altitude is calculated for the star at the moment of observation. The calculated true altitude of the star is $19°23'11.''9$. The calculated refraction correction for that altitude is $00°02'24.''3$. What altitude will be measured when the star is observed if these calculated values are correct?

(A) $19°20'47.''6$
(B) $19°23'11.''9$
(C) $19°22'15.''5$
(D) $19°25'36.''2$

19°23'11.9" = MEAS. ALT − 00°02'24.3"

Solution: Parallax is so small as to be undetectable for observations on fixed stars. In cases where the observed altitude is known and the true altitude is calculated, the refraction correction is subtracted.

$$\text{true altitude} = \text{observed altitude} - \text{refraction correction}$$

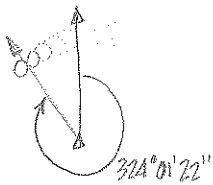
However, in this case it is the true altitude that is known and the observed altitude that must be calculated. Therefore, the refraction correction is added.

$$\begin{aligned}\text{observed altitude} &= \text{true altitude} + \text{refraction correction}\\ &= 19°23'11.''9 + 00°02'24.''3\\ &= 19°25'36.''2\end{aligned}$$

Answer is (D)

59. An observation for azimuth was taken on the trailing limb of the sun. The angle right from the azimuth mark to that limb was 324°01′22″. The semidiameter of the sun at the time of the observation was 00°15′59.″5, and the true altitude of the sun was 42°19′06.″1. If the azimuth of the sun at the moment of observation was 146°01′47.″6, what is the azimuth of the line to the azimuth mark?

(A) 181°44′26.″1
(B) 181°38′48″
(C) 178°21′12″
(D) 178°15′33.″9



Solution: The first step in solving this problem is to find the correction in azimuth for the semidiameter, sometimes known as dH.

$$dH = \frac{\text{SD}}{\cos h}$$

SD is the semidiameter, and h is the true altitude.

$$\begin{aligned}dH &= \frac{00°15'59.''5}{\cos 42°19'06.''1}\\ &= 00°21'37.''6\end{aligned}$$

Next, the angle right is increased by the semidiameter correction since the observation was made to the trailing limb.

$$324°01'22'' + 00°21'37.''6 = 324°22'59.''6$$

Finally, find the azimuth of the line to the azimuth mark.

$$\begin{aligned} \text{azimuth of the sun} &= 146^\circ 01' 47''.6 \\ \text{corrected angle right} &= \underline{-324^\circ 22' 59''.6} \\ & -178^\circ 21' 12'' \\ & \\ & -178^\circ 21' 12'' \\ & \underline{+360^\circ 00' 00''} \\ \text{azimuth of the line} &= 181^\circ 38' 48'' \end{aligned}$$

Answer is (B)

60. The semidiameter of the sun gradually changes from approximately $00^\circ 16'.3$ in the winter months to approximately $00^\circ 15'.8$ in the summer months. Which of the following statements best describes the reason for this variation?

(A) The earth reaches aphelion in the winter and perihelion in the summer.
(B) Refraction is greater in the summer than it is in the winter.
(C) Refraction is less in the summer than it is in the winter.
(D) The earth is closer to the sun in the winter than it is in the summer.

Solution: The earth reaches perihelion, which is its closest point to the sun, in January, and aphelion in July. Therefore, the sun's semidiameter appears to be largest about Jan. 3 of each year and then slowly decreases until approximately July 4.

Answer is (D)

OBSERVATIONS OF POLARIS

61. Polaris, the pole star, is found in which of the following constellations?

(A) Ursa Major
(B) Sagittarius
(C) Canis Major
(D) Ursa Minor

Solution: Polaris is the α star in the constellation Ursa Minor—the brightest star in the Little Bear. The constellation is also known as the Little Dipper.

Answer is (D)

62. The polar distance of Polaris never exceeds which of the following values?

(A) $00°30'$
(B) $00°01'$
(C) $01°00'$
(D) $00°45'$

Solution: Polaris is known as a circumpolar star because it never dips below the horizon in the northern hemisphere. It is very close to the north celestial pole, always within $1°$, and appears to rotate counterclockwise in a circular path.

Answer is (C)

63. The maximum and minimum altitudes of circumpolar stars, including Polaris, are sometimes described by which names?

(A) upper and lower elongation
(B) inner and outer elliptic
(C) upper and lower zenith
(D) upper and lower culmination

Solution: Both the maximum and minimum altitudes of Polaris occur on th meridian of the observer and are called the *upper* and *lower transit* of the star These positions of Polaris have been given the names *upper* and *lower culmina tion*.

Answer is (D)

64. Which of the stars listed is closest to the south celestial pole?

(A) Mensa
(B) α Hydrus
(C) β Chamaeleon
(D) σ Octantis

Solution: σ Octantis is far from the brightest star in its constellation, but it appears to rotate in a circle of less than 1° around the southern celestial pole. In this way, σ Octantis is similar to Polaris in the northern hemisphere.

Answer is (D)

65. The positions of Polaris at which it appears to reach the maximum east and west extents of its path around the northern celestial pole are called by which names?

(A) eastern and western extension
(B) upper and lower elongation
(C) eastern and western parallactics
(D) eastern and western elongation

Solution: Eastern and western elongation are the names given to the farthest east and west positions of the path of Polaris around the north celestial pole. However, it is not correct to say that the altitude of Polaris when it reaches elongation is equal to that of the north celestial pole. Polaris at elongation is slightly higher than the elevation of the north celestial pole from an observer's point of view.

Answer is (D)

66. The altitude of Polaris corrected for refraction provides a good approximation of which of the following?

(A) right ascension
(B) declination
(C) longitude of the observer
(D) latitude of the observer

Solution: Since the path of Polaris never departs from the north celestial pole by more than 1°, its true altitude provides a fairly good approximation of the latitude of the observer.

Answer is (D)

67. Which of the following statements describes advantages of observing Polaris for azimuth at elongation?

(A) The calculation of the azimuth of the star is simplified.
(B) The effect of refraction is minimized.
(C) The star moves less than 5 sec in azimuth within 3 min of elongation.
(D) Both A and C are true.

Solution: Polaris appears to move almost vertically for approximately 3 min before and after elongation, making the observation of the star convenient. The calculation of the azimuth at elongation is also much more convenient than at other positions. If a disadvantage exists for observations of Polaris at elongation, it is that the position may be reached at inconvenient times.

Answer is (D)

68. Polaris is seen at its eastern elongation by an observer at the position

$$\begin{aligned} \mathrm{N}\phi &= 39°49'49'' \\ \mathrm{W}\lambda &= 104°59'12'' \end{aligned}$$

The declination of Polaris is 89°13′16″. What is the azimuth of the star?

(A) 00°59′56″
(B) 01°00′45″
(C) 00°48′19″
(D) 01°00′51″

Solution: The calculation of the azimuth of Polaris at elongation is simpler than it is in any other position and may be found using the expression

$$\sin \mathrm{Az} \text{ (azimuth of Polaris at elongation)} = \frac{\cos\delta}{\cos\phi}$$

δ is the declination of Polaris and ϕ is the latitude of the observer.

$$\begin{aligned} \sin \mathrm{Az} &= \frac{\cos 89°13'16''}{\cos 39°49'49''} \\ &= 0.0177015 \\ \mathrm{Az} &= 01°00'51'' \end{aligned}$$

Answer is (D)

69. At the moment of the eastern or western elongation of Polaris, which of the angles of the PZS triangle is 90°?

(A) the hour angle
(B) the parallactic angle
(C) the codeclination
(D) the coaltitude

Solution: The eastern and western elongations of Polaris occur when the parallactic angle is 90°. The hour circle and the vertical circle through the star are perpendicular to one another.

Answer is (B)

The following three problems refer to the data given.

An observer occupied a station with the following latitude and longitude.

$$\mathrm{N}\phi = 39°49'46''$$
$$\mathrm{W}\lambda = 104°59'10''$$

The eastern local hour angle of Polaris at the moment of observation was 118°46′35″, and its declination was +89°12′37″. The observer backsighted a reference mark and measured a horizontal angle right of 258°19′58″ to Polaris.

70. What is the azimuth of Polaris at the moment of observation?

(A) 00°55′42″
(B) 00°53′12″
(C) 00°53′47″
(D) 00°56′30″

Solution: The hour angle formula is convenient for this calculation.

$$\tan \mathrm{Az} = \frac{\sin t}{\cos\phi \tan\delta - \sin\phi \cos t}$$

ϕ is the latitude, δ is the declination, and t is the hour angle.

$$\begin{aligned}
\tan \mathrm{Az} &= \frac{\sin 118°46'35''}{(\cos 39°49'46'')(\tan +89°12'37'') - (\sin 39°49'46'')(\cos 118°46'35'')} \\
&= \frac{0.876505}{(0.767954)(72.54722) - (0.640504)(-0.481392)} \\
&= \frac{0.876505}{55.7129 - (-0.308334)} \\
&= \frac{0.876505}{56.021295} \\
&= 0.015646 \\
\mathrm{Az} &= 00°53'47''
\end{aligned}$$

Answer is (C)

71. What is the azimuth of the line from the observer to the reference mark?

(A) 100°46′15″
(B) 102°33′55″
(C) 90°56′12″
(D) 102°33′49″

Solution: The angle right from the reference mark to Polaris was 258°19′58″, and the azimuth of Polaris was 00°53′47″ at the moment of observation.

$$258°19'58'' - 00°53'47'' = 257°26'11''$$

The angle to the left from north is 257°26′11″. Therefore, the azimuth will be this angle less 360°.

$$\text{azimuth from observer to reference mark} = \begin{array}{r} 360°00'00'' \\ -257°26'11'' \\ \hline 102°33'49'' \end{array}$$

Answer is (D)

72. What was the altitude of Polaris at the moment of observation?

(A) $39°49'46''$
(B) $39°26'45''$
(C) $38°50'21''$
(D) $37°14'52''$

Solution: The formula used to find the altitude of a star is

$$\sin h = \sin\phi \sin\delta + \cos\phi \cos\delta \cos t$$

h is the altitude, ϕ is the latitude, δ is the declination, and t is the local hour angle.

$$\begin{aligned}\sin h &= (\sin 39°49'46'')(\sin 89°12'37'') \\ &\quad + (\cos 39°49'46'')(\cos 89°12'37'')(\cos 118°46'35'') \\ &= 0.640444 + (-0.005095) \\ &= 0.635348 \\ h &= 39°26'45''\end{aligned}$$

Answer is (B)

OBSERVATION FOR LATITUDE

73. The lower limb of the sun was observed at the moment of transit on the observer's meridian, which is at apparent noon. The declination of the sun at the moment of observation was $+5°41'07''$. The refraction correction was $00°01'05''$ and the parallax correction was $00°00'06''$. The semidiameter was $00°15'40''$. The measured altitude of the sun was $45°15'42''$. What is the latitude of the observer?

(A) $49°42'57''$
(B) $40°17'03''$
(C) $51°42'11''$
(D) $50°10'44''$

Solution: The angular distance from the celestial equator to the observer's zenith is equal to the observer's latitude.

$$\begin{aligned}
\text{measured altitude of sun at transit} &= 45°15'42'' \\
\text{correction for parallax} &= \underline{+00°00'06''} \\
& 45°15'48'' \\
\text{correction for refraction} &= \underline{-00°01'05''} \\
& 45°14'43'' \\
\text{correction for semidiameter} &= \underline{+00°15'40''} \\
\text{true altitude of sun's center} &= 45°30'23''
\end{aligned}$$

$$\begin{aligned}
\text{altitude of sun} &= 45°30'23'' \\
\text{sun's declination} &= \underline{-05°41'07''} \\
\text{height of celestial equator} &= 39°49'16''
\end{aligned}$$

$$\begin{aligned}
\text{angular distance from horizon to zenith} &= 90°00'00'' \\
\text{height of celestial equator} &= \underline{-39°49'16''} \\
\text{observer's latitude} &= 50°10'44''
\end{aligned}$$

Answer is (D)

THE EQUATION OF TIME

74. Which of the following expressions correctly describes the difference between mean solar time and true solar time at any particular instant?

(A) the right ascension of the mean sun
(B) sidereal time
(C) the right ascension of the true sun
(D) the equation of time

Solution: The length of a true solar day—the time required for the earth to make a complete revolution on its axis with respect to the true sun—varies throughout the year. The length of a mean solar day—the time required for a complete revolution with respect to the imaginary mean sun—is constant. The difference between the time at any instant as measured by the mean sun and as measured by the true sun is called the *equation of time*.

Answer is (D)

75. Which of the following values is never exceeded by the equation of time?

(A) 00 hr 01 min
(B) 00 hr $16\frac{1}{2}$ min
(C) 00 hr $07\frac{1}{4}$ min
(D) 00 hr 05 min

EQUATION OF TIME NEVER EXCEEDS 16 1/2 MINUTES

Solution: The equation of time reaches its maximum negative value in February and its maximum positive value in November, but neither value exceeds 00 hr $16\frac{1}{2}$ min.

Answer is (B)

76. An hour measured by the apparent motion of the true sun, which is sundial time, varies with the time of year. An hour measured by the imaginary mean sun is constant throughout the year. Which of the following reasons account for this effect?

(A) The orbit of the earth around the sun is elliptical. Therefore, the apparent motion of the true sun is more rapid when the earth is closer to the sun and slower when the earth is farther away.
(B) The apparent motion of the true sun is along the ecliptic and the motion of the imagined mean sun is along the celestial equator. Therefore, the mean sun moves steadily in terms of right ascension, while the true sun does not.
(C) The tidal forces of the moon and other planets of the solar system cause the rate of the rotation of the earth on its axis to vary significantly during the year. This variation causes the apparent motion of the true sun to vary.
(D) Both A and B are true.

Solution: According to Kepler's law of planetary motion, the earth moves more rapidly along its orbital path at perihelion and more slowly at aphelion. While this accounts for some variation in the motion of the true sun through the sky, it is not the largest factor. The true sun appears to move along the ecliptic due to the inclination of the earth's axis with respect to its orbital plane with the

sun. Therefore, it does not move in right ascension at a constant rate as does the mean sun. The mean sun is imagined to move steadily in right ascension along the celestial equator.

Answer is (D)

ZONE TIME

77. Which of the following phrases correctly defines the meaning of the abbreviation LMT?

(A) local meridian time
(B) lunar median time
(C) longitudinal meridian time
(D) local mean time

Solution: The term *local mean time* describes the time reckoned by the mean sun at a particular meridian. The local mean time will be different for any two places on the earth that are not precisely north or south of one another.

Answer is (D)

78. If the following local hour angles are expressed in terms of hours, minutes, and seconds, which formula correctly represents local mean time?

(A) LMT = LHA (of the true sun) + equation of time
(B) LMT = LHA (of the true sun) + 6 hr
(C) LMT = LHA (of the mean sun) + 12 hr
(D) LMT = LHA (of the mean sun)

Solution: Noon, or 12 hr, is determined in LMT to be the moment of the upper transit of the mean sun on the meridian of a particular place. Since the local hour angle of the mean sun would be considered zero at the same moment, 12 hr

should be added to the local hour angle of the mean sun to find the local mean time. Similarly, the addition of 12 hr to the local hour angle of the mean sun at any moment would yield the local mean time.

Answer is (C)

79. Which of the following would reckon noon from the upper transit of the mean sun on the Greenwich meridian?

(A) Greenwich sidereal time
(B) 7th zone time
(C) international atomic time
(D) universal time

Solution: The Greenwich, or zero, meridian remains the foundation for timekeeping around the world, but the term *Greenwich mean time* (GMT) has been largely supplanted by the term *universal time*. Midnight on the Greenwich meridian is considered 0 hr universal time.

Answer is (D)

80. There are four time zones across the contiguous 48 states of the U.S. Which of the following correctly represents the central meridians of those time zones?

(A) 60°, 70°, 80°, and 90°
(B) 75°, $87\frac{1}{2}$°, 100°, and $112\frac{1}{2}$°
(C) 80°, 85°, 90°, and 95°
(D) 75°, 90°, 105°, and 120°

Solution: Beginning with the 75th meridian of west longitude, the central meridians of the time zones across the 48 states increase by 15°. This incrementation of 15° is due to the rate of the movement of the mean sun, which is 15° of longitude per hour.

Answer is (D)

84. An aircraft left Chicago at 17 hr 00 min Central Standard Time (CST) and arrived in Tucson at 17 hr 23 min Mountain Standard Time (MST). The central meridian for CST is 90°Wλ. The central meridian for MST is 105°Wλ. To the nearest minute, what was the elapsed flight time of the aircraft?

(A) 02 hr 15 min
(B) 01 hr 10 min
(C) 00 hr 23 min
(D) 01 hr 23 min

Solution: One way to solve this problem is to convert the arrival time of the aircraft to the zone time of the zone from which it departed.

$$\begin{aligned} \text{MST} = &\quad 17 \text{ hr } 23 \text{ min} \\ &\underline{+01 \text{ hr } 00 \text{ min}} \\ \text{CST} = &\quad 18 \text{ hr } 23 \text{ min} \end{aligned}$$

The difference between the departure and arrival times is found by subtraction.

$$\begin{aligned} \text{CST of the arrival} = &\quad 18 \text{ hr } 23 \text{ min} \\ \text{CST of the departure} = &\quad \underline{-17 \text{ hr } 00 \text{ min}} \\ \text{elapsed flight time} = &\quad 01 \text{ hr } 23 \text{ min} \end{aligned}$$

Answer is (D)

85. An aircraft left Boston at 14 hr 06 min EST and arrived in Des Moines at 16 hr 37 min CST. The central meridian for EST is 75°Wλ. The central meridian for CST is 90°Wλ. To the nearest minute, what was the elapsed flight time of the aircraft?

(A) 02 hr 16 min
(B) 03 hr 31 min
(C) 01 hr 12 min
(D) 03 hr 10 min

Solution: Converting the arrival time to its equivalent in EST makes the computation of the elapsed flight time more convenient.

$$\begin{aligned} \text{CST} = \quad & 16 \text{ hr } 37 \text{ min} \\ & \underline{+01 \text{ hr } 00 \text{ min}} \\ \text{EST} = \quad & 17 \text{ hr } 37 \text{ min} \end{aligned}$$

$$\begin{aligned} \text{arrival time in EST} = \quad & 17 \text{ hr } 37 \text{ min} \\ \text{departure time in EST} = \quad & \underline{-14 \text{ hr } 06 \text{ min}} \\ \text{elapsed flight time} = \quad & 03 \text{ hr } 31 \text{ min} \end{aligned}$$

Answer is (B)

86. A solar observation was taken at $105°00'00''$Wλ on July 10. The local hour angle to the true sun at the moment of the observation was $46°12'42''$ and the equation of time was -00 hr 06 min 22 sec. The mean sun is ahead of the true sun. To the nearest minute, what was the UT at the moment of the observation?

(A) 10 hr 31 min
(B) 22 hr 11 min
(C) 10 hr 11 min
(D) 22 hr 31 min

Solution: First, express the hour angle of the true sun in hours, minutes, and seconds.

$$\frac{46°12'42''}{15} = 03 \text{ hr } 04 \text{ min } 51 \text{ sec}$$

Add 12 hr to this time since the LHA is measured from the upper transit of the sun.

$$\begin{aligned} & 03 \text{ hr } 04 \text{ min } 51 \text{ sec} \\ & \underline{+12 \text{ hr } 00 \text{ min } 00 \text{ sec}} \\ & 15 \text{ hr } 04 \text{ min } 51 \text{ sec} \end{aligned}$$

Since the equation of time is negative, the mean sun is ahead of the true sun by 00 hr 06 min 22 sec.

$$\begin{array}{rr} & 15 \text{ hr } 04 \text{ min } 51 \text{ sec} \\ & \underline{+00 \text{ hr } 06 \text{ min } 22 \text{ sec}} \\ \text{MST of the observation} = & 15 \text{ hr } 11 \text{ min } 13 \text{ sec} \end{array}$$

The result is in MST since the observation was taken on the 105th meridian. The UT is found by adding 7 hr.

$$\frac{105°\text{W}\lambda}{15} = 7 \text{ hr}$$

$$\begin{array}{r} 15 \text{ hr } 11 \text{ min } 13 \text{ sec} \\ \underline{+07 \text{ hr } 00 \text{ min } 00 \text{ sec}} \\ 22 \text{ hr } 11 \text{ min } 13 \text{ sec} \end{array}$$

To the nearest minute, the UT was 22 hr 11 min.

Answer is (B)

SIDEREAL TIME

87. A local sidereal day is the interval of time between successive upper transits of the First Point of Aries on the observer's meridian. Which of the following transits also occurs at intervals of 24 sidereal hours?

(A) successive upper transits of the true sun
(B) successive upper transits of the mean sun
(C) successive lower transits of Saturn
(D) successive upper transits of Aldebaran

Solution: All of the bodies listed except the star Aldebaran have variable right ascensions. Aldebaran's angular distance from the First Point of Aries, which is its right ascension, is virtually constant. This is the case with all of the so-called fixed stars. Twenty-four sidereal hours pass between successive transits of Aldebaran.

Answer is (D)

88. An observation was made at the moment of upper transit of the star Antares. If the right ascension of the star was 247°10′45″, what was the sidereal time at the moment of the observation?

(A) 07 hr 31 min 17 sec
(B) 19 hr 31 min 17 sec
(C) 04 hr 28 min 43 sec
(D) 16 hr 28 min 43 sec

Solution: The right ascension of a star is measured from the First Point of Aries opposite the direction of the apparent rotation of the celestial sphere. The right ascension of Antares is more than 180°, so the sidereal time must be greater than 12 hr when the star reaches upper transit. The exact sidereal time can be found by dividing the right ascension of the star by 15, since the First Point of Aries is considered to move 15° every sidereal hour.

$$\frac{247°10'45''}{15} = 16 \text{ hr } 28 \text{ min } 43 \text{ sec}$$

Answer is (D)

89. There are approximately $365\frac{1}{4}$ solar days in a year. What is the number of sidereal days in a year?

(A) 365 sidereal days
(B) $364\frac{1}{4}$ sidereal days
(C) $366\frac{1}{4}$ sidereal days
(D) $304\frac{1}{2}$ sidereal days

Solution: The time required for the earth to orbit the sun is precisely 365.24222 solar days. The same interval of time in sidereal days requires exactly one more revolution of the earth upon its axis, or 366.24222 sidereal days.

Answer is (C)

90. Which of the following intervals of mean solar time corresponds to the interval between the sidereal times of 02 hr 12 min 15 sec and 06 hr 45 min 32 sec?

(A) 04 hr 34 min 02 sec
(B) 04 hr 33 min 17 sec
(C) 04 hr 32 min 32 sec
(D) 04 hr 31 min 47 sec

Solution: First, determine the sidereal interval between the two sidereal times given.

$$\begin{array}{r} 06 \text{ hr } 45 \text{ min } 32 \text{ sec} \\ \underline{-02 \text{ hr } 12 \text{ min } 15 \text{ sec}} \\ 04 \text{ hr } 33 \text{ min } 17 \text{ sec} \end{array}$$

Next, the ratio between the number of mean solar days in the year and the number of sidereal days in the year provides a convenient factor for finding the equivalent solar interval.

$$\frac{365.24222}{366.24222} = 0.997269567$$

Multiply this value by the sidereal interval to find the mean solar interval.

$$(04 \text{ hr } 33 \text{ min } 17 \text{ sec})(0.997269567) = 04 \text{ hr } 32 \text{ min } 32 \text{ sec}$$

Answer is (C)

91. Which of the following intervals of sidereal time corresponds to the mean solar time interval of 24 hr 00 min 00 sec?

(A) 23 hr 56 min 03 sec
(B) 24 hr 00 min 00 sec
(C) 24 hr 05 min 27 sec
(D) 24 hr 03 min 57 sec

Solution: Use the ratio developed in the solution to Problem 90 to make this conversion.

$$\left(\frac{366.24222}{365.24222}\right)(24 \text{ hr } 00 \text{ min } 00 \text{ sec}) = 24 \text{ hr } 03 \text{ min } 57 \text{ sec}$$

Answer is (D)